Response to the Associate Editor and Reviewers

Manuscript: *FlowMOP: An Automated Flow Cytometry Time, Debris, and Doublet Removal Tool*

The Associate Editor and reviewers raised several overlapping concerns. To avoid repeating the same response while ensuring that every point is addressed, closely related comments are reproduced verbatim and answered together below. A complete comment-coverage index is provided at the end of this response.

We thank the Associate Editor and reviewers for their considered feedback and believe that the requested amendments have significantly improved the manuscript. We hope that the revised manuscript now addresses the Associate Editor's and reviewers' concerns.

Data Leakage: Correction to the Comparative Benchmark

During revision, we identified a data-leakage error in the original competitor benchmark: the source-label field retained for scoring had inadvertently been available to PeacoQC and FlowCut. The synthetic FCS files deliberately retained `SampleIDInt` so that event-level source truth remained attached for scoring, but the original PeacoQC and FlowCut benchmark paths did not remove this numeric field from the channels available for quality assessment. FlowMOP did not have this issue because its fluorescence-channel selection explicitly excluded channel names containing `sample`, together with Time and scatter channels.

We reran the full benchmark with the source label excluded from all quality-control inputs and used it only for scoring. The correction reinforced rather than reversed the comparative findings. FlowMOP had higher sensitivity than FlowCut for Segment at both tested bin sizes and for the 5000-event Bimix and Trimix benchmarks. At 5000 events, FlowMOP also had higher specificity than both competitors across Segment, Bimix, and Trimix; at 2000 events, it retained higher specificity than PeacoQC in Bimix and Trimix without a significant sensitivity disadvantage and had higher sensitivity and specificity than FlowCut in Segment. Figure 2, its violin distributions, p-values, significance brackets, and all reported comparisons now use these corrected outputs.

This correction is independent of the specific reviewer comments addressed below and is reported for transparency.

Associate Editor's General Assessment

> Novel tools to expedite flow cytometry data analysis are always welcome, and I commend the authors for their effort in introducing these new methods. However, as currently submitted, the manuscript requires major revision. The reviewers have identified numerous areas requiring improvement, and I encourage the authors to address their comments carefully and comprehensively. In addition to the reviewers' critiques, I offer several observations of my own that I believe will strengthen the manuscript's suitability for publication.

Response: The revisions are organized around the algorithmic motivation, comparative mechanism, computational benchmarking, interpretation of expert evaluations, methodological clarity, and the limits of the current approach. Detailed responses follow.

Combined Comment 1 — Algorithmic Motivation, Novelty, Related Tools, and Validation Framing

Raised by: Associate Editor; Reviewer 2, Comments 1, 6, and 8.

Associate Editor — Algorithmic Motivation and Theoretical Justification

> The authors acknowledge the existence of other algorithms that perform substantially similar analyses to the components of FlowMop, yet they do not adequately motivate what deficiencies in the existing algorithms their new approach is designed to address. A compelling framing would follow a structure such as: we demonstrated that PeacoQC and FlowCut fail under realistic conditions X, Y, and Z; upon analysis, we determined this is attributable to specific aspects of how those algorithms operate; accordingly, we designed our algorithm to overcome these limitations. No such explanation is currently provided, nor is there a clear justification for why the particular algorithmic choices were made.

Reviewer 2, Comment 1

> In the abstract, the authors claim "FlowMOP is the first automated approach capable of debris cleaning". However, there are methods like GateNet, 10.1002/cyto.a.20531, and UNITO that can do automatic gating and probably can be used to gate out debris. There is even a review for this purpose (10.3389/fimmu.2015.00380).

Reviewer 2, Comment 6

> The introduction did not introduce available related tools.

Reviewer 2, Comment 8

> The authors claimed, "Conversely, this paper seeks to compare performance not only through traditional human-expert defined standards, but also the development of bespoke data generated explicitly for pre-processing validation." In fact, both simulation data and read data are commonly used for new method validation.

Response

We now frame FlowMOP as a training-free preprocessing workflow that combines automated time, debris, and doublet removal. We revised the Introduction to distinguish automated population-gating tools, time-quality-control tools, and FlowMOP's intended role as an integrated preprocessing workflow. We removed the claim that FlowMOP is the first automated approach capable of debris cleaning. The revised Abstract describes alternative preprocessing algorithms such as FlowCut and PeacoQC, while the Introduction discusses them as time-dependent quality-control approaches and distinguishes GateNet, UNITO, and FATE as broader population-identification or representation-learning approaches.

We have also revised the description of the three FlowMOP modules. Time gating is based on positive-population fluorescence summaries across acquisition order, debris gating derives an FSC-A threshold from cross-parameter positive-population structure, and doublet gating uses FSC-A/FSC-H and SSC-A/SSC-H ratio structure. These algorithmic choices are now distinguished from computational implementation choices.

In response to Reviewer 2's observation that simulated and real data are both commonly used in method validation, we now describe the synthetic samples specifically as event-labelled technical controls developed to provide objective ground truth for flow cytometry preprocessing tasks.

Changes made

- We removed the claim that FlowMOP is the first automated approach capable of debris cleaning and added the following clarification:

> "Several tools automate cytometry population identification, including deep-learning approaches [7,8], the model-based flowClust method [9], the neural-network method GateNet [10], and the bivariate-segmentation framework UNITO [11], while FATE uses representation learning to produce generalized flow cytometry embeddings [12]; automated gating approaches are reviewed elsewhere [13]. Conversely, FlowMOP is a Python-based, training-free preprocessing workflow that combines automated time-gating, debris removal, and doublet exclusion in a single headless tool. FlowCut and PeacoQC provide the most direct comparison for its time-gating component."

- We reframed the synthetic-data contribution as follows:

> "The synthetic datasets provide event-level labels for estimating sensitivity and specificity in preprocessing tasks where real ground truth is otherwise difficult to define."

- We added a concise description of the three algorithmic components to the Figure 1 legend:

> "A) FlowMOP selects valid parameters using positive-peak detection, generates time-binned fluorescence summaries, applies two smoothing resolutions, and performs robust outlier rejection across acquisition order. B) FlowMOP applies the same valid-parameter check, derives a candidate FSC-A threshold from each eligible parameter's positive events, and uses the median candidate as the final FSC-A gate. C) FlowMOP identifies doublets from inflection points in FSC-A/FSC-H and SSC-A/SSC-H ratio histograms, with a fixed-ratio fallback."

Location: Introduction; Results, "Algorithmic design," "Time Gating," "Debris Gating," and "Doublet Gating"; Figure 1 legend.

Combined Comment 2 — Computational Benchmarking and Within-Sample Parallelisability

Raised by: Associate Editor; Reviewer 1, Comment 1; Reviewer 2, Comment 7.

Reviewer 1, Comment 1 — Technical Infrastructure and Performance Claims

> The adoption of a Dask-based Python framework is a highly commendable architectural choice. Flow cytometry analysis has historically under-utilized distributed compute resources, and providing a modern, scalable alternative to R-based packages is a significant contribution to the "Big Data" era of spectral cytometry. However, the manuscript currently lacks empirical data to support the claims of improved runtime or memory efficiency, while other contemporary algorithms explicitly highlight their scalability for large datasets.
>
> Recommendation: To make a compelling case for FlowMOP's adoption, the authors could include benchmarking figures (e.g., execution time and peak memory usage)

comparing FlowMOP against PeacoQC and FlowCut across datasets of increasing file count and/or size (~10^3 to ~10^7 events or some [Event x Parameter] matrix varying each). Another point of comparison may be in the distributed compute scenario vs. more traditional local compute resources.

Example Performance Benchmarking Table:

Dataset Size (Events)	Metric	FlowMOP (Python/Dask)	PeacoQC (R)	FlowCut (R)
10^4	Execution Time (s)			
	Peak RAM (MB)			
10^5	Execution Time (s)			
	Peak RAM (MB)			
...etc.				

Associate Editor — Algorithmic vs. Implementation Advantages

Reviewer 1 raises several points regarding the use of the Dask framework, and I would like to extend that line of inquiry. Specifically, I would ask the authors to clarify whether there is something inherent to the FlowMop algorithm that makes it particularly well-suited to parallelization via Dask — as distinct from the implementation itself. This distinction matters considerably. Given current development tools, a competent programmer with AI tooling could likely port the existing FlowCut or PeacoQC code — neither of which is especially lengthy — to a Dask-based parallel implementation in a matter of hours. If that is the case, it raises the question of whether a substantial portion of FlowMop's reported advantage derives from the implementation rather than from any intrinsic algorithmic innovation, which would significantly diminish the novelty of the contribution. I would encourage the authors to focus their comparative analysis on algorithmic advantages, particularly in cases where existing algorithms present genuine structural barriers to parallelization.

Reviewer 2, Comment 7

The authors should introduce "Dask" before using the "Dask-based framework" in the introduction. Please keep in mind, most readers probably have no experience using Python at all. I use Python a lot myself. Still, I am not familiar with this "Dask" thing. Also, the Dask-based design is about calculation efficiency, instead of accuracy. Readers will be more interested in which algorithm is used for automatic gating, instead of how to speed up calculation. The authors did not really introduce the algorithm for gating in the introduction. But they use a whole paragraph on the Dask-based design.

Response

We evaluated computational performance using matched 36-channel FCS inputs containing 10,000, 100,000, 300,000, 1,000,000, and 2,000,000 events. FlowMOP, PeacoQC, and FlowCut were run on the same clone-based inputs. Optional plotting, reporting, and output generation were disabled where supported, and FlowMOP's debris and doublet modules were disabled so that the shared time-gating task was timed fairly. Each condition included one warm-up and three measured repeats. Runtime and peak resident memory are reported as mean ± SD in Table 1. At 2,000,000 events, FlowMOP was approximately 2.5-fold faster than PeacoQC and 3.5-fold faster than FlowCut while using approximately 32% and 45% of their respective peak RAM.

The benchmark used local non-distributed execution. Testing active Dask execution showed that its overhead did not improve this workload, so Dask was inactive for the reported benchmark and removed from the manuscript's novelty framing.

Among the three evaluated decision structures, only FlowMOP is inherently suited to independent within-sample channel parallelisation. Once shared acquisition bins have been defined, each eligible channel independently establishes its fluorescence reference, produces a fixed-shape time-bin summary, and applies smoothing and MAD filtering. Cross-channel coordination occurs only when the resulting bin flags are combined in the final parameter vote. This creates coarse, regular channel-level tasks with a single final reduction.

PeacoQC must instead reconcile peak configurations across bin-channel combinations before its MAD, Isolation Tree, and consecutive-bin decisions. FlowCut combines segment- and channel-level statistics through adjacent-segment and file-wide comparisons, contiguous-region decisions, and an optional flagged-file rerun. Both methods can process separate files in parallel, and some internal suboperations could be parallelised, but their complete within-sample decision pipelines contain intermediate dependencies that prevent the same independent channel-wise reduction without changing their decision rules. Reimplementation in Python or another scheduling framework would not remove those structural dependencies.

FlowMOP's within-sample parallelisability is therefore an inherent property of its decision structure rather than its programming language. The reported runtime benchmark used local non-distributed execution, so it measures the performance of the tested implementations rather than an empirical distributed-computing speedup.

Changes made

- We removed Dask from the manuscript's novelty and accuracy framing and specified the benchmark configuration as follows:

  > "For fair timing of the shared time-gating task, FlowMOP was run using local non-distributed execution, with debris and doublet removal disabled and annotated output FCS writing disabled. PeacoQC and FlowCut were run with optional plotting, reporting, and output generation disabled where supported."

- We state FlowMOP's contribution and identify the most direct time-gating comparators:

  > "FlowMOP is a Python-based, training-free preprocessing workflow that combines automated time-gating, debris removal, and doublet exclusion in a single headless tool. FlowCut and PeacoQC provide the most direct comparison for its time-gating component."

- We retained a concise operational description in the Methods:

  > "FlowMOP accepts `.csv`, `.fcs`, and Parquet files. It reduces event-level mea-

surements into time-bin and channel summaries, applies smoothing and robust outlier detection, and combines the resulting flags through parameter voting."

- We placed the architectural comparison and its relationship to the computational benchmark in the Discussion:

  "These measurements were obtained using local non-distributed execution and therefore demonstrate the performance of the tested implementations, not a distributed-computing speedup."

  "Among the three evaluated decision structures, only FlowMOP exposes an inherently channel-parallel within-sample computation. After shared acquisition bins are defined, each eligible fluorescence channel independently establishes its reference, produces a fixed-shape time-bin summary, and applies smoothing and MAD filtering; cross-channel coordination occurs only in the final parameter vote. PeacoQC instead performs data-dependent peak identification and reconciliation across bin-channel combinations, while FlowCut applies adjacent-segment, contiguous-region, file-wide, and conditional rerun decisions [5,6]. Their suboperations and separate files can be parallelised, but their complete within-sample decision pipelines cannot be expressed as the same independent channel-wise reduction without changing their decision rules. Porting either implementation to another language or scheduling framework would therefore not remove these structural dependencies. This structural distinction, together with FlowMOP's earlier data reduction, fewer full-data passes, and smaller intermediate state, is consistent with its lower runtime and memory use in our benchmarks, although the benchmark does not independently establish causation."

- We added matched runtime and peak-memory testing across five event counts:

  "Computational scalability was evaluated using clone-based real-FCS scaling. A representative FCS file was subsampled/replicated to matched event counts of 10,000, 100,000, 300,000, 1,000,000, and 2,000,000 events while preserving the original 36-channel structure. FlowMOP, PeacoQC, and FlowCut were run on the same generated inputs for each size."

- We reported the principal computational result as follows:

  "The computational benchmark demonstrates that FlowMOP provides speed gains with lower peak RAM usage at larger event counts. This is most evident from 300,000 events onward, where FlowMOP had both the fastest mean runtime and lowest peak memory use. At 2,000,000 events, FlowMOP was approximately 2.5-fold faster than PeacoQC and 3.5-fold faster than FlowCut, while using approximately 32% of PeacoQC's peak RAM and 45% of FlowCut's peak RAM."

Location: Methods, "Computational scalability benchmark"; Results, "Algorithmic design" and "Computational scalability"; Table 1; Discussion, "Synthetic Sample Time Gating."

Combined Comment 3 — Downstream Biological Validation, Dataset Breadth, and the Cost of Errors

Raised by: Associate Editor; Reviewer 1, Comments 3 and 4; Reviewer 2, general assessment and Comment 10.

Associate Editor — Downstream Biological Impact

> Alternatively — or in addition — the authors could evaluate the downstream biological impact of each gating strategy. By substituting FlowMop, FlowCut, and PeacoQC gates for the time and debris gating steps while retaining the human gating strategy for all other steps, one could directly assess whether differences in automated gating performance have any meaningful effect on the biological conclusions drawn from the data. Ultimately, the relevant question for end users is not whether an algorithm removes a few percent more or fewer debris events, but whether doing so materially affects downstream analysis.

Reviewer 1, Comment 3 — Biological Dataset Breadth

> Methodological Robustness: Utilizing a single expert per tissue type limits the statistical power of the ranking methodology. It prevents the reader from seeing the intra-tissue consensus—how consistently multiple experts on the same tissue agree on the rank-order of methods used on a specific sample. What it currently highlights is how a single tissue expert has a unique perspective of the correct gating for a given tissue and inter-operator gating variability; not algorithmic gating superiority. Other improvements could be adding different processing conditions (e.g. fixation) or more variation in tissue used (e.g. include human PBMC or other species' tissues).

Reviewer 1, Comment 4 — Discussion of Error "Costs" and Sensitivity Trade-offs

> The manuscript notes that PeacoQC exhibits higher sensitivity but lower specificity compared to FlowMOP. This trade-off is central to clinical discovery and warrants a deeper discussion.
>
> Discussion Point: The authors should explicitly address the biological "cost" of these errors. For instance, is it more detrimental to the study's integrity to leave in a small transient artifact or to accidentally remove a rare, clinically significant biological population?

Reviewer 2 — Difficult Biological Samples and Tumour Validation

> The authors developed An Automated Flow Cytometry Time, Debris, and Doublet Removal Tool. This can be a useful tool for cytometry. However, the dataset used in the paper are relatively "easy" cases. I recommend the authors to challenge the FlowMOP with more difficult but common samples like digested tumor sample. Performance against human expert is needed.

Reviewer 2, Comment 10

> The study did not validate the FlowMOP on tumor datasets, which is a common sample type for flow cytometry but difficult to perform debris removal. I highly suggest the authors to include a tumor dataset, even if the performance is not optimal, the readers can know whether the method can be applied in their sample or not.

Response

We elected to conduct further biological validation, and therefore added PBMC samples from eight donors and three human liver-tumour samples. These analyses quantify PBMC population recovery and biological composition and tumour-population recovery.

We agree with Reviewer 1 that the original single-expert-per-tissue design cannot establish within-tissue expert consensus or support claims of algorithmic superiority. We therefore present it as an expert-preference evaluation, retain the complete analysis in the Supplementary Information, and report its principal comparative findings in the main Results, as detailed in Combined Comment 6. To broaden the biological validation, we added the PBMC and tumour datasets described here; we did not add a separate fixation or cross-species experiment.

For the PBMC analysis, we compared the time-gating methods while holding the downstream expert-defined singlet, debris, Live-cell, and lineage coordinates fixed. We also evaluated the debris and doublet modules separately against matched expert inputs in which only the relevant cleaning step differed, and compared the complete Time + Debris + Doublet workflow with the expert-combined workflow.

Live CD45+ was reported as a standalone reference endpoint. Applying it as an additional parent gate excluded many low-scatter events before the lineage endpoints were quantified and therefore obscured the effects of debris removal. To expose those effects directly, we did not include CD45+ as the parent of the lineage populations. B cells were defined as Live CD3−CD19+ events, T cells as Live CD3+CD19− events, and NKT cells as Live CD19−CD3+CD56+ events. We report counts and frequencies normalized to their corresponding matched Raw values (Raw = 1); before normalization, B-, T-, and NKT-cell frequencies were calculated relative to Live cells. All analyses used paired PBMC samples from the same eight donors. Direct workflow comparisons used two-sided paired *t*-tests, while Raw comparisons used one-sample *t*-tests against 1 on the Raw-normalized values, equivalent to paired comparisons with the matched Raw values; tests were Holm-adjusted separately within endpoint and outcome metric.

For time gating, no evaluated B-, T-, or NKT-cell frequency differed significantly among the methods (all adjusted p values were 0.420 or greater). Relative to Expert Manual, FlowMOP retained 11.1–12.7 percentage points more B, T, and NKT cells (all adjusted p values were 0.003 or less), FlowCut retained 12.3–13.8 points more (all adjusted p values were 0.006 or less), and PeacoQC retained 11.7–13.4 points fewer (all adjusted p values were 0.035 or less; Figure 5).

Figure 6 reports debris-only, doublet-only, and combined preprocessing in three subcolumns per population. FlowMOP generally removed fewer events overall than Expert Manual during debris cleaning. No debris count endpoint differed between the methods, and no population frequency differed except T-cell frequency, which was lower after FlowMOP cleaning (Raw-normalized mean 1.013 versus 1.069; adjusted p = 0.037). Doublet cleaning produced no significant count or frequency difference between FlowMOP and Expert Manual. When Time, Debris, and Doublet preprocessing were combined, no count endpoint differed; B-cell frequency was the only frequency that differed and was lower after FlowMOP cleaning (0.877 versus 1.022; adjusted p = 0.049). These results indicate broad agreement between human- and FlowMOP-mediated pregating. Supplementary Figure S8 contains representative debris and doublet preprocessing projections, with gate outlines shown only where the corresponding preprocessing gate was applied.

In the tumour analysis, the B- and T-cell endpoints are population recovery relative to Raw, not frequency or composition within the cleaned output, because their source counts use the original total-event denominator. No statistically detectable differences were observed between

Manual and FlowMOP for the evaluated recovery endpoints in these three samples (Figure 7). Given n = 3, these nonsignificant results do not establish equivalence.

The biological data alone cannot establish whether the lower retention by Expert Manual and PeacoQC reflects more sensitive artifact removal or whether the higher retention by FlowMOP and FlowCut reflects more specific preservation of valid events. Considering that PeacoQC demonstrated poorer specificity than FlowMOP and FlowCut in the synthetic datasets, together with the relatively blunt nature of human manual time gating, we believe that the higher retention is more likely to reflect specific preservation of valid events than insufficient artifact removal.

The trade-off between sensitivity and specificity reflects competing biological risks rather than a purely statistical optimization. A more permissive gate may protect genuine or rare populations while allowing artifacts to remain; a more stringent gate may remove artifacts more completely while also deleting valid biological events. No balance is universally correct because the consequences of each error depend on the intended downstream analysis. We therefore report both sensitivity and specificity and interpret them alongside their effects on population recovery and composition. Importantly, FlowMOP had the strongest overall combined sensitivity-specificity profile in the source-labelled synthetic time-gating benchmarks: its gains were not achieved simply by exchanging one error type for the other.

These biological analyses better illustrate FlowMOP's capabilities and reinforce the findings demonstrated in the synthetic datasets. Figure 7A additionally shows representative Manual and FlowMOP Time, Debris, and Doublet preprocessing plots.

Changes made

- We added a biological validation using PBMC samples from eight donors, with matched-Raw frequencies and counts for Live CD45+, B, T, and NKT cells. Figure 5 reports time-cleaning representatives (A), frequencies (B), and counts (C). Figure 6 reports the combined Time + Debris + Doublet representative (A) and the debris-only, doublet-only, and combined frequency (B) and count (C) comparisons. Supplementary Figure S8 contains the debris and doublet preprocessing projections; gate outlines appear only where the corresponding preprocessing gate was applied.

- We found no direct count difference between FlowMOP and Expert Manual in the debris, doublet, or combined comparisons. No doublet frequency differed between methods; the only direct frequency differences in Figure 6 were debris-only T-cell frequency (adjusted p = 0.037) and combined B-cell frequency (adjusted p = 0.049).

- We added a tumour validation comprising three tumour samples, with paired Raw, Manual, and FlowMOP comparisons of Live CD45+ cell-count recovery and B- and T-cell population recovery. Figure 7 displays the matched gating plots and downstream comparisons. No statistically detectable differences were observed between Manual and FlowMOP for these recovery endpoints, but n = 3 does not support an equivalence claim.

- We added representative Manual and FlowMOP Time, Debris, and Doublet preprocessing plots (Figure 7A).

- We acknowledged that the single-expert-per-tissue rankings cannot estimate within-tissue consensus, retained the complete ranking analysis in the Supplementary Information, and added its principal comparative findings to the main Results. We broadened the validation with human PBMC and tumour datasets but did not add a separate fixation or cross-species experiment.

- We added a direct discussion of competing error costs:

> "The trade-off between sensitivity and specificity reflects competing biological risks rather than a purely statistical optimization. A more permissive gate may protect genuine or rare populations while allowing artifacts to remain; a more stringent gate may remove artifacts more completely while also deleting valid biological events. No balance is universally correct because the consequences of each error depend on the intended downstream analysis. In this context, FlowMOP's synthetic time-gating results are notable because its performance gains were not achieved simply by exchanging one error type for the other. FlowMOP had the strongest overall combined sensitivity-specificity profile across the tested conditions."

Location: Methods, "Human PBMC biological-validation sample preparation," "Biological-validation analysis," and "Tumour biological-validation analysis"; Results, "Biological validation" and "Tumour biological validation"; Discussion, "Biological validation" and "Other remarks"; Figures 5–7; Figure S8.

Combined Comment 4 — Gating Mechanism, Smoothing, and Parameter Fairness

Raised by: Associate Editor; Reviewer 1, Comment 5; Reviewer 2, Comment 20.

Associate Editor — Algorithmic Motivation and Theoretical Justification

> More troublingly, the authors themselves appear uncertain about why their algorithm performs better in certain cases. In the Discussion, they write that a performance difference "may be attributed to FlowMop's 'smoothing' implementation." This speculation is unsatisfying — can the authors not test this directly by modifying the smoothing implementation and evaluating whether the results change accordingly? While presenting cherry-picked examples in which one algorithm outperforms another has illustrative value, a rigorous justification of the theoretical underpinnings of the algorithmic design is essential. Without it, readers cannot reasonably assess which sample types and experimental conditions are expected to favor FlowMop over existing alternatives.

Reviewer 1, Comment 5 — Parametric Sensitivity Analysis

> To strengthen the claim of "superiority," it is essential to demonstrate that the results are not merely a function of suboptimal default settings for the competing algorithms.
>
> Recommendation: A supplemental figure showing a sensitivity analysis—where the parameters of PeacoQC and FlowCut are tuned across a range—would provide a more "apples-to-apples" comparison and confirm that FlowMOP's performance gains are algorithmic rather than parametric.

Reviewer 2, Comment 20

> The authors say, "Finally, FlowMOP also novelly employs the use of smoothing at multiple resolutions to ensure multiple types of abherrations are detected". What are the "multiple types of abherrations"? please clarify.

Response

We now seek to address the mechanisms underlying FlowMOP's superior performance under the primary fixed comparison settings in the tested Segment, Bimix, and Trimix scenarios. Our hypotheses arose from the different signals used by the methods and from the pattern observed in the original benchmark. FlowMOP uses acquisition order to define time bins but bases its removal decision on fluorescence summaries of globally defined positive populations; Time itself is excluded from the quality variables. FlowCut, by contrast, can respond to local acquisition-density structure. This distinction was particularly relevant because the Segment files retained stronger source-linked Time-density structure and showed the clearest FlowCut performance loss, whereas Time was approximately normalized in the constructed Bimix and Trimix files. We therefore hypothesised that FlowCut's relative loss arose, at least in part, from sensitivity to local acquisition rate rather than to the source-linked fluorescence defect itself.

To test this hypothesis, we conducted a matched Time-only mechanism analysis. This involved changing only the Time channel while preserving fluorescence, scatter, source labels, and event order across 30 source-labelled Bimix, Trimix, and Segment files. We found that FlowMOP and PeacoQC were both unchanged under source-linked and random Time warping, whereas FlowCut's removal behaviour changed when local Time density was altered. The clearest loss occurred in Segment inputs under source-linked Time warping.

PeacoQC's documentation identifies acquisition-bin size as a trade-off between the accuracy of within-bin density estimation, the number of bins available for evaluating signal stability, and the number of events affected when a bin is removed [5]. The Bimix and Trimix benchmarks contain short, interspersed source-defined intervals. FlowMOP's stronger performance is therefore consistent with its temporal summarisation being better matched to these brief intervals.

In the regenerated primary benchmark, FlowMOP had higher sensitivity than FlowCut in both Segment settings and in the 5000-event Bimix and Trimix benchmarks. At 5000 events, FlowMOP also had higher specificity than both competitors across Segment, Bimix, and Trimix. In the smaller mixed-source benchmarks, FlowMOP retained higher specificity than PeacoQC without a significant sensitivity disadvantage.

Consequently, the earlier smoothing attribution has been removed. The matched Time-only benchmark directly supports the distinction between FlowMOP's fluorescence-population-summary decision and FlowCut's response to acquisition-density changes when fluorescence is unchanged. PeacoQC's documented bin-size trade-off provides the relevant context for its fixed-setting comparison with FlowMOP.

We thank the reviewers for prompting this direct examination of smoothing. We tested a range of smoothing settings across all synthetic time-gating inputs while holding all other FlowMOP settings fixed (Table S4). No smoothing provided the best specificity, but at the cost of sensitivity. Inspection showed that the additional target-source events removed under smoothing lay in distribution shoulders, where target and non-target events were difficult to distinguish reliably. We therefore retained smoothing as the conservative default: it avoids treating ambiguous shoulder events as confidently valid and provides higher sensitivity, while accepting the corresponding specificity trade-off. Among the smoothed settings, 0.01,0.05 provided the strongest balance and was selected as the default. We reran FlowMOP across all Figure 2 inputs and regenerated Figure 2 using these outputs.

All algorithms were compared using documented recommended, default, or automatically selected settings, including fixed FlowMOP parameters, to reflect typical unsupervised use. We did not perform extensive parameter tuning for FlowCut or PeacoQC because the original method descriptions do not provide dataset-specific guidance for how such tuning should be performed. We therefore restrict our conclusions to the fixed comparison settings and the behaviours directly tested by the matched Time-only benchmark.

The two smoothing resolutions are now described as complementary spline fits applied to the same time-bin fluorescence-summary series before MAD filtering. A bin can be detected through either smoothing pass; the manuscript no longer relies on the imprecise phrase "multiple types of aberrations."

Changes made

- We removed the earlier smoothing attribution and added the following matched comparison:

  > "To test the effect of flow-rate disturbances aligned with or independent of source-linked fluorescence changes, we altered only the Time channel while leaving fluorescence, scatter, source labels, and event order unchanged (Fig. S4). FlowMOP and PeacoQC were unchanged under both source-linked and random Time warping. In contrast, FlowCut's sensitivity and specificity shifted after Time-only perturbation."

- We clarified the function of the two smoothing resolutions as follows:

  > "Two spline smoothing values, one small and one larger (current default 0.01,0.05), are applied to the returned time-bin series before median absolute deviation (MAD) filtering."
  >
  > "The two smoothing resolutions target both shorter and more sustained deviations, while parameter voting limits removal driven by isolated noisy channels in higher-dimensional panels."

- We expanded the smoothing analysis across all 173 primary synthetic time-gating inputs. Supplementary Table S4 reports the tested range and identifies 0.01,0.05 as the strongest-performing smoothed setting on the balanced comparison.

- We reran FlowMOP using the selected 0.01,0.05 setting and regenerated the Figure 2 violin distributions, p-values, and significance brackets from these outputs.

- We clarified the scope of the fixed-setting comparison:

  > "The primary comparison used recommended or automatically selected settings, including fixed FlowMOP parameters, to reflect typical unsupervised use."

- We added the complete mechanism-benchmark methods and results as Figure S4.

Location: Methods, “Time-only acquisition-rate mechanism benchmark” and “MAD-smoothing ablation and default selection”; Results, “Time Gating” and “Synthetic Time Gating Benchmark”; Figure 2; Figure S4; Table S4; Discussion, “Synthetic Sample Time Gating.”

Combined Comment 5 — Parameter Voting and the Ten-Parameter Threshold

Raised by: Reviewer 2, Comments 18 and 21.

Reviewer 2, Comment 18

> The authors say ”when >10 parameters are present, bins are rejected if they are flagged by two or more parameters”. Why “10” is used as threshold here?

Reviewer 2, Comment 21

> The authors say, “parameter voting to ensure that higher dimensional sample sets are not overly degraded.” It would be nice to show how bad the overly degradation will be is parameter voting is not applied. It is a little a bit concern that events with aberrant signal in one fluorescence are not detected, even in a big panel.

Response

The revised manuscript identifies the ten-parameter cutoff as an empirical operational threshold rather than a mathematically derived boundary.

The rationale for requiring two channel flags in panels with more than ten eligible parameters is conservative. Biological datasets have a non-zero probability of isolated channel-specific outliers, and the opportunity for such outliers increases with panel dimensionality. Requiring corroboration from a second parameter reduces the risk that a single noisy channel removes otherwise valid biological events. The corresponding trade-off is that an abnormality confined to one fluorescence channel may be retained in a high-dimensional panel. We did not add a separate voting-ablation analysis and have avoided presenting the rule as universally optimal.

Changes made

- We added the following rationale and limitation:

  > “For panels with more than 10 parameters, FlowMOP requires two or more parameters to flag a bin before rejection. This empirical safeguard reduces false-positive removal caused by isolated noisy channels in high-dimensional panels, although an aberration confined to one channel may consequently be retained (Figure 1A).”

- We did not add a voting-ablation experiment and do not present the threshold as universally optimal.

Location: Results, “Time Gating.”

Combined Comment 6 — Expert Rankings, Terminology, Visualisation, and Bayesian Modelling

Raised by: Associate Editor; Reviewer 1, Comment 3; Reviewer 2, Comment 16.

Associate Editor — Expert Preference Rankings and Statistical Interpretation

My other major concern pertains to the substantial space — both in the main manuscript and supplemental materials — devoted to preferential rankings by four experts across multiple sample types and algorithms, analyzed using a Bayesian statistical framework. Setting aside the question of whether all experts can reasonably be considered equally expert across all sample types, and the limited statistical power afforded by four data points, I find the interpretation of these rankings fundamentally ambiguous.

As expected, experts generally preferred human-gated data over automated tools — but this finding is difficult to interpret in isolation. Forcing experts to rank outputs necessarily produces an ordering, but that ordering conflates two very different scenarios: one in which all automated results are unacceptable and the other in which all automated results are perfectly adequate, with human gating simply preferred at the margin. Similarly, even if FlowMop was ranked above the other automated tools, the reader has no basis for judging whether this difference reflects a meaningful improvement in analytical utility or merely a marginal statistical preference with no practical consequence.

I would suggest that the authors replace or supplement the ranking analysis with a simpler, more interpretable quality scale — for example: 1 = excellent gating; 2 = adequate and suitable for analysis; 3 = inadequate for analysis but not terrible; 4 = unacceptable. This approach would convey absolute quality rather than relative preference and would be more informative to readers.

Reviewer 1, Comment 3 — Refinement of Human Subjective Rankings

While the subjective rankings provide an interesting window into expert perspectives on automated gating, the presentation and terminology in this section require significant refinement:

Nomenclature: In computer science, the term “benchmarking” typically implies an objective, repeatable measure of performance in specific scenarios. In the context of subjective human preference, “Comparison” or “Evaluation” is a more accurate/appropriate term. This sentence, “FlowMOP’s outputs in these samples were benchmarked against a set of human experts”, should more precisely say that FlowMOP's outputs were “compared against expert gating”.

Methodological Robustness: Utilizing a single expert per tissue type limits the statistical power of the ranking methodology. It prevents the reader from seeing the intra-tissue consensus—how consistently multiple experts on the same tissue agree on the rank-order of methods used on a specific sample. What it currently highlights is how a single tissue expert has a unique perspective of the correct gating for a given tissue and inter-operator gating variability; not algorithmic gating superiority. Other improvements could be adding different processing conditions (e.g. fixation) or more variation in tissue used (e.g. include human PBMC or other species' tissues).

Data Visualization: The ranking scale (7 = Best, 1 = Worst) is possibly counter-intuitive and should be reversed to follow standard ordinal ranking conventions. The row axis title in the grid figure should be updated to “Gate Provided By” to clearly disambiguate between the creator of the gate (e.g., “Expert 1”) and the individual performing the ranking (“ Expert”).

The scatter plot in Panel B adds limited value; replacing it with an “Average Score” column at the end of the grid would be cleaner. This statement is oddly contorted to be true, but is misconstruing the data: “However, it is of note that in 4/5 and 7/9 datasets in the debris and doublets tasks respectively, FlowMOP was not the least preferred, indicating superiority to at least one human benchmarker (Fig. 6A, 7A)”. Another interpretation is that FlowMOP had the lowest average score in Fig. 7A/B and had the fewest tissue cases where experts deemed it as the best result or even better-than-average traditional gating.

Reviewer 2, Comment 16

There is only a “Bayesian Modelling” section under the “Non-synthetic samples” section in the Method. The Bayesian Modelling section described the method to rank performance of human experts, FlowMOP, FlowCut and PeacoQC. It is not about how “Non-synthetic samples” are generated at all. Also, the purpose of the Bayesian Modelling is well described in the beginning of this subsection, which makes it difficult to follow.

Response

Forced rankings measure relative preference rather than objective event-level accuracy. We therefore replaced “benchmarking” with “expert comparison” or “expert evaluation,” removed the contorted “not least preferred” interpretation, and now state directly how FlowMOP ranked relative to the human and automated gates.

The evaluation produced distinct results across the three cleaning modules. For time gating, FlowMOP had the best mean rank among the automated methods and was preferred to FlowCut (BF = 5.39, P = 84.3%) and PeacoQC (BF = 12.10, P = 92.4%). Human gates generally ranked higher for debris and doublet removal, although FlowMOP was competitive in several tissue-specific comparisons: it ranked first for debris removal in mouse blood, third for debris removal in human liver and mouse skin, and second for doublet removal in human liver (Figs. S5-S7). We now report these principal findings and statistics in the main Results; the complete dataset-level rankings and mean ranks are shown in Figures S5-S7, with statistical comparisons reported in the Supplementary Results.

The revised Discussion considers these results alongside the other validation approaches. The preference for FlowMOP over the automated time-gating comparators agrees with its combined sensitivity-specificity performance in the synthetic benchmarks. The more variable debris and doublet rankings are discussed in relation to their dependence on sample-specific scatter distributions. Taken together, the results indicate that FlowMOP-generated gates can be comparable in acceptability to human gating across many use cases.

The evaluation was designed as a relative-ranking exercise, with each dataset assessed by an expert familiar with that sample type. We did not retrospectively replace this with an absolute-quality scale because doing so would require repeating the assessment under a different protocol. Instead, we retain the ranking analysis and report its outcomes directly, alongside the source-labelled synthetic controls and the new PBMC and tumour analyses. The reviewer's request for broader biological datasets, including human PBMCs, and the Associate Editor's request for downstream biological endpoints are addressed together in Combined Comment 3.

The visual summaries now include an Average Score column within each ranking grid rather than a separate panel, and the row axis is labelled "Gate Provided By." The displayed scale now follows the standard convention that rank 1 is best. This display correction did not change the underlying ranking order or Bayesian analysis.

The Bayesian model is retained because the observed outcome is an ordinal ranking. The Plackett–Luce model retains the complete ordering and represents uncertainty in relative preference without treating differences between adjacent ranks as equal-interval continuous measurements. We have clarified that it estimates relative preference only. The subsection is presented as the statistical analysis of the expert-ranking data rather than as a method for generating non-synthetic samples.

Changes made

- We defined the scope of the expert-ranking analysis as follows:

  > "The expert-ranking analysis was used to summarize relative preferences among gates generated by FlowMOP, comparator algorithms, and human operators; it was not treated as an absolute measure of gating adequacy [16]."

- We revised the Supplementary Results terminology and interpretation as follows:

  > "FlowMOP's outputs in these samples were compared with expert-provided gates using a forced-ranking preference task. These rankings measure relative expert preference and should not be interpreted as an absolute measure of gating adequacy."

- We retained the Plackett–Luce analysis and clarified that it models ordinal preference rankings rather than absolute gating quality.

- We moved the response concerning dataset breadth and downstream biological validation to Combined Comment 3 and cross-reference it here rather than treating the expert rankings as biological validation.

- We retained the full expert-ranking analysis in the Supplementary Information and added its principal comparative findings and statistics to the main Results, with their implications discussed alongside the synthetic and biological-validation findings.

- We regenerated Supplementary Figures S5–S7, changed the row-axis title to "Gate Provided By," displayed rank 1 as best, and replaced the separate summary panel with an Average Score column in each grid.

- We revised the Figure S5 caption as follows:

  > "Figure S5. Expert preference rankings for time gates provided by four human experts, FlowMOP (black border), FlowCut, and PeacoQC across nine datasets. Rows indicate the gate provider, and the final column shows the mean rank across datasets. Rank 1 indicates greatest preference; therefore, a lower average score indicates greater preference. Abbreviations: DRG, dorsal root ganglion; CNS, central nervous system."

- We clarified the scoring direction in the Supplementary Results and figure captions as follows:

> "Rank 1 indicates the greatest preference."

Location: Results, "Expert preference evaluation"; Discussion; Conclusion; Supplementary data, "Supplementary expert preference evaluation," including "Supplementary methods" and "Supplementary results"; Supplementary Figures S5–S7.

Combined Comment 7 — Dataset Scale and Complexity

Raised by: Associate Editor.

Associate Editor — Smaller Point

> In the Introduction, the authors state that sophisticated analysis software has contributed to datasets of increasing size and complexity. It is not clear to me how analysis software makes debris or time gating more complex, particularly when preprocessing is predominantly scatter-based. Do the authors perhaps mean that derived parameters from imaging cytometry increase pre-processing complexity? Similarly, the claim that "datasets exceed human capabilities" presumably refers to the sheer number of files rather than the complexity of individual datasets — this should be stated explicitly. It is also not apparent from the manuscript how the tools described here depend on data complexity per se.

Response

The revised Introduction distinguishes study scale from the intrinsic complexity of an individual preprocessing gate. It now refers explicitly to increasing numbers of files, events, and measured parameters and explains that this scale makes repeated manual preprocessing time-consuming and often impractical. It no longer implies that analysis software itself necessarily makes a debris or time gate more complex.

Changes made

- We clarified the distinction between study scale and individual-gate complexity as follows:

  > "Modern flow cytometry studies can contain increasing numbers of files, events, and measured parameters [1]. The large scale of these data makes repeated manual preprocessing time-consuming and often impractical, potentially amplifying variability between operators. Reproducible automated preprocessing is therefore valuable for large studies."

Location: Introduction.

Combined Comment 8 — Acquisition Settings and Newer Scatter Configurations

Raised by: Reviewer 2, Comments 13 and 28.

Reviewer 2, Comment 13

> The voltage set of the cytometer significantly impact the resolution of debris. If the voltage is relatively low with bad resolution, can FlowMOP perform the automatic

> task well? Also, please describe the voltage setting in the data generation section in the Method.

Reviewer 2, Comment 28

> The Xenith flow cytometry form Thermo Fisher provides 405 FSC/SSC, 488 FSC/SSC, and polar 488 FSC/SSC. This provides more flexible approach to gate target population. How can the FlowMOP adapt to this kind of flow data? It would be nice to discuss this in the Discussion.

Response

The benchmarking relied on real, existing datasets, each acquired using experiment-specific, biology-driven antibody panels and instrument settings. Consequently, acquisition voltage/gain settings differed among datasets and were not controlled variables in this study. As with the antigen–fluorochrome combinations, these settings are therefore not material to the preprocessing comparisons reported here, and a single voltage/gain specification would not describe the datasets used. We have clarified this scope in the Methods.

Acquisition settings nevertheless determine whether the scatter signals needed by the debris and doublet modules are adequately resolved. FlowMOP requires appropriate acquisition voltage/gain settings; if relevant signals are poorly resolved or saturated, the lost information cannot be recovered and reliable cleaning cannot be guaranteed. FlowMOP currently expects users to identify the scatter channels used by the workflow; it has not been validated systematically across 405-nm, 488-nm, and polar 488-nm FSC/SSC configurations on instruments with multiple scatter measurements. Future versions could assess multiple scatter-channel pairs and select or combine the pair with the clearest debris/doublet separation.

Changes made

- We clarified the dataset-specific acquisition settings in the Methods:

  > "The benchmarking uses real, existing datasets, each acquired using experiment-specific, biology-driven antibody panels and instrument settings. Consequently, the antigen–fluorochrome combinations and acquisition voltage/gain settings differ among datasets and were not variables under evaluation."

- We retained the following combined acquisition/scatter limitation:

  > "FlowMOP requires appropriate acquisition voltage/gain settings; if relevant signals are poorly resolved or saturated, the lost information cannot be recovered and reliable cleaning cannot be guaranteed. FlowMOP currently expects users to identify the scatter channels used by the workflow; it has not been validated systematically across 405-nm, 488-nm, and polar 488-nm FSC/SSC configurations on instruments with multiple scatter measurements. Future versions could assess multiple scatter-channel pairs and select or combine the pair with the clearest debris/doublet separation."

Location: Methods, "Synthetic time source-sample preparation"; Discussion, "Other remarks."

Combined Comment 9 — Temporal Artifacts and Synthetic Time-Sample Design

Raised by: Associate Editor; Reviewer 2, Comments 3, 4, and 14.

Associate Editor — Smaller Point

> Regarding the bimix and trimix data used to simulate realistic fluorescence fluctuations: I am uncertain how these mixtures reproduce the kinds of flow rate variations that occur during actual experiments. If I understand correctly, the simulated fluctuations were subtle enough that human reviewers could not reliably detect them. I would ask the authors to clarify whether this represents a clinically or experimentally common problem — in my experience, flow rate deviations typically manifest as clear departures at the beginning or end of an acquisition, with mid-acquisition instabilities being relatively uncommon.

Reviewer 2, Comment 3

> The authors should explain the concept of "temporal artifacts" in the abstract, given this is the core issue to be addressed in the study.

Reviewer 2, Comment 4

> There are many terms in the abstract "(concatenated and mixed time perturbations, high-debris mixtures, and CFSE/CTV co-labeled doublets)". I am not sure whether it is a good idea to put them here without explanation, considering unexplained term might add confusion to the readers. Anyway, the authors should introduce these terms in the introduction, which is missing.

Reviewer 2, Comment 14

> The "Generation of Synthetic Time Samples" section is difficult to understand. This section is important for readers to understand the validation results. I recommend the authors use a illustration to show these 3 synthetic manners. Also, please explain why these three manners are designed in this way? So that the readers can follow the text.

Response

The revised text explains the practical relevance of the Bimix and Trimix designs. It defines a "microblockage" operationally as a short, self-resolving mid-acquisition disturbance that produces a localized fluorescence shift; the term does not imply that a physical obstruction was directly observed. Existing work supports the substance of this phenomenon even though it uses broader terminology. flowAI documented flow-rate surges interspersed throughout acquisitions and their association with signal-intensity variation [4]. PeacoQC describes temporary acquisition shifts caused by slow sample uptake or a clog and notes that such problems can be difficult to detect manually [5], while FlowCut describes manual identification and removal of transient acquisition problems as time-consuming and subjective [6].

Because the affected intervals can be short and their events interspersed with otherwise plausible events, they are difficult to recognize reliably by eye and impractical to exclude using a series of manual gates. We use "microblockage" for this subtle, self-resolving acquisition failure mode that a time-quality-control method should be able to detect. Estimating its prevalence across experiments would require a separate study.

The Bimix and Trimix samples model the observable fluorescence consequence in this operational definition rather than reproducing or proving a physical obstruction. This distinction is important. The synthetic samples can appear acceptable by eye, but their retained source labels reveal short intervals containing events from an intentionally perturbed fluorescence source. Under the benchmark definition, those events should be excluded even when visual inspection alone would not identify a defensible manual gate. The apparent visual normality of these files is therefore the reason an event-labelled benchmark is needed.

The downstream biological-validation analyses are presented together in Combined Comment 3. That evidence provides an independent biological context for the technical benchmark; it is cross-referenced here rather than repeated. Together, the synthetic and biological analyses motivate an evaluation that includes both readily visible sustained artifacts and visually subtle transient microblockages.

We also defined temporal artifacts, simplified the terminology, and added Figure S1. We clarified that the primary synthetic samples model fluorescence changes across acquisition order without flow-rate disturbances, because rate changes without corresponding fluorescence changes should not prompt event exclusion. Flow-rate effects were tested separately, either aligned with source-linked fluorescence changes or independently of them. FlowMOP was unchanged in both conditions, whereas FlowCut's removal behavior shifted (Figure S4).

Changes made

- In the Abstract, we incorporated the definition into the methodological summary:

  > “Methodologically, FlowMOP identifies temporal artifacts—acquisition-dependent deviations in event quality or fluorescence signal—via parameter-wise peak checks, bin-level fluorescence summaries across acquisition time, and robust outlier rejection.”

- In the Methods, we added the following clarification:

  > “Following acquisition, events from these real, source-labelled FCS files were computationally recombined to construct three time-gating benchmark designs. Their construction is illustrated in Supplementary Figure S1, and the dataset compositions are reported in Supplementary Table S1.”

  The wet-lab PBMC protocol is now presented separately under “Synthetic time source-sample preparation,” followed by “Computational construction of Segment, Bimix, and Trimix datasets.” This distinguishes preparation of the real source samples from post-acquisition benchmark construction.

  > Here, we use “microblockage” operationally to denote a short, self-resolving mid-acquisition disturbance that produces a localized fluorescence shift; the term does not imply that a physical obstruction was directly observed. Segmented samples model sustained changes, whereas Bimix and Trimix model the observable fluorescence consequence in this operational definition by introducing short source-defined fluorescence shifts during acquisition. They do not recreate or establish the physical mechanism itself. The Bimix and Trimix files can appear acceptable on visual inspection because the altered intervals are short and interspersed with otherwise plausible events; however, the retained source labels identify the intentionally perturbed events that should be excluded under the

benchmark definition.

> “Flow-rate disturbances without corresponding fluorescence changes should not prompt event exclusion; these samples therefore model fluorescence changes across acquisition order without introducing flow-rate disturbances. Flow-rate effects were tested separately by altering Time either in alignment with source-linked fluorescence changes or independently of them.”

- In the Results, we added the following comparison:

> “To test the effect of flow-rate disturbances aligned with or independent of source-linked fluorescence changes, we altered only the Time channel while leaving fluorescence, scatter, source labels, and event order unchanged (Fig. S4). FlowMOP and PeacoQC were unchanged under both source-linked and random Time warping. In contrast, FlowCut's sensitivity and specificity shifted after Time-only perturbation.”

- In the Discussion, we now explain why the visually subtle synthetic cases are experimentally relevant and require an event-labelled benchmark:

> Transient acquisition disturbances are not confined to the beginning or end of a run: flow-rate surges interspersed throughout acquisition and associated signal-intensity variation have been documented [4]. PeacoQC notes that temporary acquisition problems can be difficult to detect manually [5], and FlowCut describes manual identification and removal of transient acquisition problems as time-consuming and subjective [6]. We use “microblockage” operationally for a short, self-resolving instance of this broader phenomenon that produces a localized fluorescence shift, without asserting that a physical obstruction was directly observed.
>
> “Although these synthetic samples can appear acceptable on visual inspection, the source labels show that the short altered intervals contain events from an intentionally perturbed fluorescence source and therefore should be excluded under the benchmark definition. Visual subtlety is thus a central feature of the benchmark: it demonstrates why apparent normality by eye is not sufficient ground truth.”

- We cross-reference the consolidated downstream biological-validation response in Combined Comment 3 rather than repeating its analyses here.

- We added Figure S1 with the following caption:

> “Figure S1: Construction of Segment, Bimix, and Trimix synthetic time samples. No flow-rate disturbance was introduced.”

Location: Abstract; Introduction; Methods, “Synthetic time source-sample preparation,” “Computational construction of Segment, Bimix, and Trimix datasets,” and “Time-only acquisition-rate mechanism benchmark”; Results, “Synthetic Time Gating Benchmark”; Discussion, “Synthetic Sample Time Gating”; Figures S1 and S4; Table S1. The biological-validation changes are detailed in Combined Comment 3.

Combined Comment 10 — Debris Methodology and Validation

Raised by: Reviewer 1, Comment 2; Reviewer 2, Comments 15, 22, and 24.

Reviewer 1, Comment 2 — Debris Gating Methodology

> The inclusion of an automated debris-gating method is a welcome addition, particularly as this is often omitted in other QC tools. However, the current reliance on FSC-A as the primary determinant for debris may explain the lower rankings in expert consensus (Figure 6A).
>
> Critique: While the rationale is that debris is generally "low-size," this does not account for larger morphological debris, such as large-cell clumps (e.g. tumor) or non-biological aggregates, which can occupy higher FSC/SSC ranges. Metadata-based margin removal techniques may augment this debris filtering.
>
> Recommendation: The authors should provide a more robust rationale for leaning exclusively on FSC-A or, ideally, discuss how the integration of SSC-A or Pulse Width might improve debris discrimination beyond ultra-low-size events.

Reviewer 2, Comment 15

> The authors say "For assessment of FlowMOP gating performance, 'high debris' and 'low debris' samples were synthetically combined." How are they combined? Please remember, well-described Method guarantees the transparency of the study.

Reviewer 2, Comment 22

> The FlowMOP applies an FSC-A based threshold to exclude debris. I have concern about this FSC-A based idea. Commonly, we use FSC vs. SSC scatter to gate debris, which is clear in samples like PBMC. However, the resolution of a simple FSC-A histogram will not be as good as that in a FSC vs. SSC scatter, because debris and cells will overlap in FSC-A axis. In the Figure 3, the authors show representative results which is based on simple samples, which is possible to use the FSC-A based threshold method to gate debris. However, in difficult cases, which is not very rare, this FSC-A based threshold method may perform very bad.

Reviewer 2, Comment 24

> I would like to see the debris remove rate calculated against the human expert.

Response

The revised Methods describes the debris-removal procedure more precisely. FlowMOP ultimately applies a one-dimensional FSC-A threshold, but that threshold is not obtained from a single unconditioned FSC-A histogram. The algorithm identifies eligible fluorescence parameters, examines the FSC-A structure of each parameter's positive population, derives a candidate FSC-A threshold from each eligible channel, and applies the median candidate threshold to the sample. Thus, information across eligible fluorescence channels informs the final FSC-A gate, while the applied decision remains a conservative FSC-A threshold.

The current module is not a universal debris classifier. Debris and intact cells can overlap on FSC-A, and large clumps, tumour-associated material, non-biological aggregates, or other high-scatter contaminants may not be removed reliably. We specifically considered whether SSC-A should be incorporated. SSC-A can be informative for large or internally complex debris, but large and internally complex events can also represent desirable populations that should be retained. Their interpretation varies with tissue, staining panel, cytometer configuration, acquisition settings, and the biological populations of interest. Unlike the comparatively transferable relationship between very low FSC-A and small debris, there is no broadly safe high-SSC-A rule for excluding large events.

In the present technical-control datasets, the labelled small-debris and desirable source populations overlapped substantially along SSC-A, so a fixed SSC-A threshold offered limited additional separation for the small-event removal targeted by FlowMOP. We therefore did not incorporate SSC-A into the current debris decision. Pulse-width measurements may similarly help identify some aggregates or clumps, but their availability and interpretation are instrument- and acquisition-dependent. Incorporating SSC-A or pulse width responsibly would require a wide configurable parameter range or sample-specific multivariate decision rules validated for the intended dataset, panel, instrument, and cell populations. Metadata-based margin removal is complementary rather than a substitute for debris classification: it can identify events at acquisition limits, but it does not identify all low-FSC debris or large aggregates. FlowMOP currently includes only an FSC-A maximum-value precleaning check and does not implement a generalized margin-event filter. The revised Discussion therefore defines the current scope as conservative low-FSC debris removal and presents SSC-A, pulse-width measurements, margin-event metadata, and sample-specific multivariate models as future extensions rather than universal default filters.

Figure 3 compares FlowMOP with four expert debris gates using the source-labelled controls. Figure 6 complements this comparison with total debris-gate removal and downstream biological preservation in paired PBMC samples from eight donors. Across all events, FlowMOP excluded 3.3–56.7% of the matched debris input (mean 18.3%), compared with 6.7–43.8% for Expert Manual (mean 24.8%), and therefore removed debris-gated events in every paired input. These percentages summarize total debris-gate removal and are distinct from the downstream population values plotted in Figure 6B,C. FlowMOP retained mean counts equivalent to 91.4–94.9% of matched Raw across the Live CD45+ reference, B, T, and NKT endpoints, compared with 96.3–98.8% for Expert Manual; no count endpoint differed directly between the methods. No population frequency differed from Expert Manual except T-cell frequency, which was lower after FlowMOP cleaning (Raw-normalized mean 1.013 versus 1.069; Holm-adjusted $p = 0.037$). Figure 6 reports the downstream plots, and Supplementary Figure S8 shows the representative debris and doublet preprocessing projections. The source-labelled synthetic debris benchmark separately provides the objective assessment of targeted debris removal.

The synthetic preparation is now described explicitly. Approximately equal event numbers were sampled from the high-debris and low-debris sources and concatenated into matched mixtures while retaining source labels. These source labels provide the objective basis for the reported post-cleaning proportions.

The requested human comparison is represented by the percentage-based analysis in Figure 3. FlowMOP's enrichment of the labelled low-debris component is compared directly with four expert gates, with mean percentages, variability, and paired statistical comparisons. Because the input event count differs between samples, percentage-based retention and enrichment are more directly comparable than raw removed-event counts.

FlowMOP determines its debris gate independently for each sample, whereas manual debris gating is commonly performed groupwise by applying one gate across related samples. We therefore also asked the experts to perform both groupwise and individual-sample gating. No difference was detected between these strategies for any expert (Fig. 3D, unadjusted paired t-tests, $p > 0.05$), indicating that the comparison with FlowMOP was not driven by this difference in gating strategy.

Changes made

- We described the synthetic debris mixture as follows:

  > "For assessment of FlowMOP debris-gating performance, high-debris and low-debris samples were combined by sampling approximately equal event numbers from each source and concatenating them into matched synthetic mixtures while retaining source labels for ground-truth quantification."

- We clarified the scope and derivation of the FSC-A gate as follows:

  > "FlowMOP's debris module targets small, low-FSC debris. The final gate is applied on FSC-A, but its threshold is informed by FSC-A distributions across eligible fluorescence-positive populations rather than the overall FSC-A histogram alone."
  >
  > "SSC-A may improve recognition of larger or internally complex debris, while pulse-width measurements may assist with aggregates. Accordingly, FlowMOP does not currently incorporate SSC-A or pulse-width measurements into its debris decision because broadly applicable thresholds for these features are difficult to establish across tissues, panels, instruments, and acquisition settings."
  >
  > "The median FSC-A threshold across all parameters is taken as the final FSC-A gate to be applied to the sample (Figure 1B)."

- We expanded the Discussion of SSC-A and pulse-width integration:

  > "SSC-A may improve recognition of larger or internally complex debris, while pulse-width measurements may assist with aggregates. Accordingly, FlowMOP does not currently incorporate these features because broadly applicable decision rules are difficult to establish across tissues, panels, instruments, and acquisition settings."

- We clarified that generalized metadata-based margin removal is not currently implemented beyond the FSC-A maximum-value precleaning check and is a potential complementary extension rather than a complete debris classifier.

- We added the PBMC biological-validation result showing that FlowMOP removed debris-gated events in every paired input, with variable removal across samples. The overall removal percentages are distinguished from the downstream population values plotted in Figure 6B,C. No evaluated debris count endpoint differed directly from Expert Manual; T-cell frequency was the only population frequency that differed (adjusted $p = 0.037$). We interpret this biological preservation analysis alongside the objective source-labelled debris benchmark.

- We reported the existing expert comparison as follows:

> "FlowMOP did not differ significantly from any human evaluator except Expert 4, for whom FlowMOP removed more labelled high-debris events (Bonferroni-adjusted paired t-test, p = 0.049) (Fig. 3C)."

- We clarified that FlowMOP estimates debris gates independently for each sample and reported the existing groupwise-versus-individual-sample control, for which no difference was detected for any expert (Fig. 3D).

Location: Methods, "Synthetic Debris Sample Preparation and Generation," "Human PBMC biological-validation sample preparation," and "Biological-validation analysis"; Results, "Debris Gating," "Synthetic Debris Gating Benchmark," and "Biological validation"; Discussion, "Synthetic Sample Debris and Doublet Gating" and "Biological validation"; Figures 3 and 6; Supplementary Figure S8.

Combined Comment 11 — Aggregates, Doublets, Saturation, and Validation

Raised by: Reviewer 2, Comments 5, 25, and 26.

Reviewer 2, Comment 5

> Preprocessing sometimes include aggregates removal. Is it possible for FlowMOP to handle this task?

Reviewer 2, Comment 25

> The FlowMOP creates a histogram of the FSC-A/FSC-H ratio to gate doublets. I also have a concern about this method. One unmentioned hypothesis of this algorithm is that all cells fall within the measurement range of FSC. However, it is very common that some cells exceed the FCS limit and collapsed in the edge of a FSC-A vs FSC-H scatter plot. For example, big myeloids in PBMC, if the target population is lymphocytes. Such population is expected to lead to weird histogram of the FSC-A/FSC-H ratio. This can be difficult to be addressed by the FlowMOP and are not mentioned in the study.

Reviewer 2, Comment 26

> The authors use CTV-CFSE double positive cells to represent doublets. However, there are probably CTV- CTV doublets and CFSE - CFSE doublets, which are missed in the performance evaluation. I would like to see the doublet remove rate calculated against the human expert.

Response

FlowMOP's doublet module targets events whose FSC-A/FSC-H and, where available, SSC-A/SSC-H pulse ratios distinguish them from singlets. It can remove some higher-order coincident events when those events produce separable ratio structure, as observed for triplet-like events in the technical controls, but it cannot identify every aggregate or clump from these ratios alone.

The general requirement for appropriate acquisition settings is addressed in Combined Comment 8. For the doublet module specifically, FSC-A/FSC-H and SSC-A/SSC-H ratios must remain informative. Saturated or edge-collapsed scatter measurements may therefore require acquisition review, manual intervention, or alternative pulse-shape features. We did not add a new saturation-handling algorithm.

The reviewer is also correct that CTV-CFSE double-positive events represent heterologous labelled doublets, with rare dye-transfer events as a possible exception. CTV-CTV and CFSE-CFSE doublets cannot be distinguished from their respective single-labelled populations using the dye labels alone. The technical-control design therefore gives a sensitive measure of retained known heterologous doublets but cannot establish the complete false-positive and false-negative rate for all doublet classes.

Figure 4 already compares the percentage of CTV-CFSE double-positive events removed by FlowMOP with the corresponding percentages removed by human experts. We now describe this endpoint and its same-label limitation explicitly rather than implying complete doublet ground truth.

FlowMOP also estimates its doublet gates independently for each sample rather than applying a shared gate across a group. The experts therefore performed both groupwise and individual-sample doublet gating. These strategies did not differ for three of the four experts; Expert 3 removed fewer doublets with individual-sample gating (Fig. 4C, paired t-test, $p = 0.009$). We now explain this methodological distinction and its practical implications in the Results and Discussion.

We additionally compared FlowMOP with Expert Manual doublet cleaning in paired PBMC samples from eight donors. FlowMOP produced slightly lower mean retained counts across the Live CD45+ reference, B-, T-, and NKT-cell endpoints, but no doublet count or frequency endpoint differed significantly between methods (Figure 6B,C). When Time, Debris, and Doublet preprocessing were combined, no count endpoint differed; B-cell frequency was the only frequency that differed between FlowMOP and Expert Manual (adjusted $p = 0.049$). Supplementary Figure S8 shows the representative doublet preprocessing projections.

Changes made

- We cross-reference the general acquisition-setting limitation in Combined Comment 8 and retain only the doublet-specific requirement that the relevant scatter ratios remain informative.

- We clarified the limits of the CTV/CFSE validation as follows:

  > "Because the CTV- and CFSE-labelled populations were generated separately, events positive for both dyes were interpreted as heterologous doublets; rare dye-transfer events are a possible exception. Same-label CTV-CTV and CFSE-CFSE doublets are not identifiable from these labels alone."

- We reported the comparison with human experts as follows:

  > "No statistically significant difference was detected between FlowMOP and any expert for this endpoint (Fig. 4B, paired t-test; unadjusted $p > 0.05$)."

- We clarified that FlowMOP estimates doublet gates independently for each sample and reported the existing comparison of groupwise and individual-sample expert gating (Fig. 4C).

- We added the PBMC biological-validation comparison with Expert Manual. No doublet count or frequency endpoint differed significantly between methods; in the combined workflow, B-cell frequency was the only direct frequency difference (adjusted $p = 0.049$; Figure

6B,C). Supplementary Figure S8 shows the representative doublet preprocessing projections.

- We retained the observation that FlowMOP removed a triplet-like population in the technical-control samples and restricted that finding to those samples.

Location: Methods, "Biological-validation analysis"; Results, "Doublet Gating," "Synthetic Doublet Gating Benchmark," and "Biological validation"; Discussion, "Synthetic Sample Debris and Doublet Gating," "Biological validation," and "Other remarks"; Figures 4 and 6; Supplementary Figure S8.

Combined Comment 12 — CTV/CFSE Preparation Protocol

Raised by: Reviewer 2, Comment 27.

Reviewer 2, Comment 27

> The protocol about how the CTV-CFSE samples are prepared is not clear. Are there wash steps in between? This is very important. It would be nice if the authors can provide detailed protocol as supplementary files.

Response

We agree that the original description did not provide enough information to reproduce the CTV/CFSE preparation, particularly the wash and quenching steps. We have replaced the abbreviated description with a detailed protocol specifying the digestion conditions, filtration and RBC lysis, cell numbers, centrifugation and wash steps, dye stocks and volumes, labelling and quenching conditions, 1:1 recombination of the labelled populations, viability staining, and acquisition platform.

Changes made

The revised Methods now state:

> "To generate samples with high proportions of doublets, C57BL/6 mouse spleens were injected with 1 mL digestion buffer comprising RPMI supplemented with 50 µg/mL collagenase P (Roche) and 10 µg/mL DNase I (Roche). Spleens were incubated for 20 minutes at room temperature (RT) in a further 1 mL of digestion buffer, mechanically dissociated, and incubated for a further 20 minutes at RT. Samples were passed through a 70-µm cell strainer with 10 mL FACS wash (PBS containing 2% heat-inactivated fetal bovine serum [FBS]), centrifuged (5 minutes, 500 × *g*, RT), and the supernatant discarded. Cell pellets were resuspended in 3 mL RBC lysis buffer and incubated for 3 minutes at RT, after which cells were washed twice with FACS wash and recovered by centrifugation.
>
> For dye labelling, 5 × 10^6 cells were transferred to each fresh 15-mL tube and stained with either CellTrace Violet (CTV; Invitrogen) or carboxyfluorescein succinimidyl ester (CFSE; eBioscience). Cells were centrifuged (5 minutes, 500 × *g*, 4°C), the supernatant was removed, and pellets were resuspended in 1 mL complete IMDM (cIMDM; Gibco IMDM supplemented with 10% heat-inactivated FBS, 100 U/mL penicillin, 100 µg/mL streptomycin, 2 mM L-glutamine, and 55 µM 2-mercaptoethanol). Two microlitres of 5 mM CTV or 10 mM CFSE were added to the side of the corresponding tube; samples were rapidly inverted and mixed, then

> incubated for 10 minutes at 37°C. Labelling was quenched by adding 5 mL ice-cold cIMDM and incubating for a further 5 minutes at RT. Cells were centrifuged and resuspended in cIMDM, after which 2 × 10^6 CTV-labelled cells and 2 × 10^6 CFSE-labelled cells were combined in a single sample and incubated for 30 minutes at 37°C and 5% CO2. Samples were centrifuged, the supernatant was removed, and cells were stained for 20 minutes at RT with Fixable Viability Dye eFluor 780 (Invitrogen; 1:1,000 in PBS). Cells were washed with PBS, centrifuged, resuspended in FACS wash, and acquired on a BD LSRII flow cytometer."

Location: Methods, "Synthetic Doublet Sample Preparation and Generation".

Combined Comment 13 — Missing and Duplicated Methods Information

Raised by: Reviewer 2, Comments 11 and 12.

Reviewer 2, Comment 11

> There is duplicated line in the Method "Samples were acquired on a Cytek Northern Lights 3-laser (V/B/R) spectral flow cytometer."

Reviewer 2, Comment 12

> In the "Preparation of Human PBMC Samples for Synthetic Time Benchmarking Samples" section, the authors mentioned there are many antibodies in the staining cocktail. However, they did not list which antibodies.

Response

The duplicated cytometer sentence has been removed. The benchmarking uses real, existing datasets acquired with experiment-specific, biology-driven antibody panels and instrument settings. Consequently, the antigen–fluorochrome combinations differ among datasets and are not variables under evaluation; they are not material to the preprocessing comparisons raised here. We have clarified this scope in the Methods rather than presenting a single fluorochrome list that would not describe the distinct panels used across the benchmark datasets.

Changes made

- We removed the duplicated sentence, "Samples were acquired on a Cytek Northern Lights 3-laser (V/B/R) spectral flow cytometer."
- We clarified why a single fluorochrome list is not applicable across the benchmark datasets:

  > "The benchmarking uses real, existing datasets, each acquired using experiment-specific, biology-driven antibody panels and instrument settings. Consequently, the antigen–fluorochrome combinations and acquisition voltage/gain settings differ among datasets and were not variables under evaluation."

Location: Methods, "Synthetic time source-sample preparation."

Combined Comment 14 — Precleaning Event Removal and Threshold

Raised by: Reviewer 2, Comment 17.

Reviewer 2, Comment 17

> In the "Precleaning" section of Results, how many events are being removed in these used datasets? This information can be provided as supplementary table/figure to justify the set of threshold (5%).

Response

The current implementation uses a 1% threshold, not 5%; the inconsistent earlier value has been corrected. FlowMOP checks the number of events at the maximum FSC-A value and removes those maximum-valued events only when they exceed 1% of the sample.

We calculated this step directly from the passed_lod channel, treating values below 0.5 as removed. The rule did not activate in PBMC samples from any of the eight donors. In the three tumour samples, it removed 3,472–9,206 events, corresponding to 2.39–3.05% of each input. The complete sample-level counts and percentages are reported in Supplementary Table S3.

The 1% cutoff is an arbitrary operational safeguard, not an empirically optimized biological threshold. Its activation depends on acquisition scaling and voltage/gain settings, and the observed activation rates are dataset-specific and should not be generalized across instruments.

Changes made

- We corrected and defined the threshold as follows:

  > "FlowMOP first checks the input file for events at the limit of detection, defined here as events at the maximum FSC-A value for that sample. If the number of events at this maximum exceeds a threshold (default 1%), FlowMOP removes these maximum-valued events. Otherwise, it retains all values."

- We added the following operational rationale and dataset-specific qualification:

  > "The 1% cutoff is an arbitrary operational safeguard, not an empirically optimized biological threshold. In the biological-validation inputs, the rule did not activate in any of the eight PBMC samples, whereas it removed 3,472–9,206 events, or 2.39–3.05%, in the three tumour samples (Table S3). Activation depends on acquisition scaling and voltage/gain settings, and these observed activation rates are dataset-specific and should not be generalized across instruments."

- We added Supplementary Table S3 with the input events, events removed, percentage removed, and threshold-activation status for every biological-validation input.

Location: Results, "Precleaning"; Supplementary Table S3.

Combined Comment 15 — Figure 1 Axes and Annotations

Raised by: Reviewer 2, Comments 19 and 23.

Reviewer 2, Comment 19

> The Figure 1A missing many axis titles. Also, there are tow subplots after time-binned median fluorescence. What are these two subplots are not clearly annotated.

Reviewer 2, Comment 23

> The Figure 1B also misses many axis titles. The Figure 1C misses y-axis title.

Response

Figure 1 remains a conceptual schematic, and its axes are not treated as quantitative measurements. The revised legend explains the two smoothing resolutions in Figure 1A, identifies the debris and doublet-ratio quantities in Figures 1B and 1C, and states that the schematic fluorescence-intensity and signal-strength axes use arbitrary units.

Changes made

- We revised the Figure 1 legend as follows:

> "Figure 1. Conceptual schematics depicting FlowMOP's time-gating (A), debris-gating (B), and doublet-gating (C) methods; plotted fluorescence intensity and signal-strength axes are schematic and use arbitrary units. A) FlowMOP selects valid parameters using positive-peak detection, generates time-binned fluorescence summaries, applies two smoothing resolutions, and performs robust outlier rejection across acquisition order. B) FlowMOP applies the same valid-parameter check, derives a candidate FSC-A threshold from each eligible parameter's positive events, and uses the median candidate as the final FSC-A gate. C) FlowMOP identifies doublets from inflection points in FSC-A/FSC-H and SSC-A/SSC-H ratio histograms, with a fixed-ratio fallback."

Location: Figure 1 legend.

Combined Comment 16 — Abstract, Introduction, and Overall Manuscript Structure

Raised by: Reviewer 2, general assessment and Comments 2 and 9.

Reviewer 2 — General Assessment (Manuscript Structure)

> Also, the manuscript is not well structured. Careful edition by experienced researcher is needed to increase the readability of the manuscript.

Reviewer 2, Comment 2

> The structure of abstract is kind of odd. The first paragraph should focus on background instead of providing a conclusion.

Reviewer 2, Comment 9

> The introduction is missing an end paragraph for summary.

Response

The manuscript has been considerably restructured and rewritten. The Abstract now begins with the general preprocessing problem and introduces FlowMOP directly. Temporal artifacts are then defined within the methodological summary before the validation design and principal findings are presented. We also added a concluding Introduction paragraph that summarizes FlowMOP's intended role and previews the time-gating comparison, debris and doublet validation, expert evaluation, and computational benchmark. Considering that the paper has been considerably restructured, we hope that the new format and writing address the reviewer's concerns.

Changes made

- We reordered the Abstract so that it begins with the study context:

  > “Flow cytometry now generates high-parameter datasets whose scale and variability challenge manual preprocessing, leading to subjectivity and poor reproducibility. Here, we introduce FlowMOP, a Python-native framework that automates three major preprocessing steps—time-gating, debris removal, and doublet exclusion.”

- We added the following study summary to the end of the Introduction:

  > “Here, we introduce FlowMOP, an automated preprocessing tool for time-gating, debris removal, and doublet exclusion. We compare time-gating performance with PeacoQC and FlowCut, evaluate debris and doublet removal using synthetic ground-truth datasets and expert comparison, and benchmark computational scalability.”

- We reorganized and edited the manuscript so that the motivation, algorithmic design, technical benchmarks, biological validation, expert evaluation, and limitations are presented in a clearer progression.

Location: Abstract; final paragraph of the Introduction; structure and writing throughout the manuscript.

Associate Editor's Closing Request

> Finally, I ask that the authors also provide thorough and complete responses to all comments raised by the other reviewers.

Comment Coverage Index

Associate Editor

Associate Editor point	Response location
General assessment	Associate Editor's General Assessment
Algorithmic motivation and theoretical justification — existing alternatives and design rationale	Combined Comment 1
Algorithmic motivation and theoretical justification — smoothing mechanism	Combined Comment 4
Algorithmic versus implementation advantages	Combined Comment 2
Expert preference rankings, downstream biological impact, and statistical interpretation	Combined Comments 3 and 6
Smaller point — dataset scale and complexity	Combined Comment 7
Smaller point — Bimix/Trimix and experimental relevance	Combined Comment 9
Request for complete responses	Associate Editor's Closing Request

Reviewer 1

Reviewer 1 comment	Response location
Comment 1 — Technical infrastructure and performance claims	Combined Comment 2
Comment 2 — Debris gating methodology	Combined Comment 10
Comment 3 — Refinement of human subjective rankings and dataset breadth	Combined Comments 3 and 6
Comment 4 — Error costs and sensitivity trade-offs	Combined Comment 3
Comment 5 — Parametric sensitivity analysis	Combined Comment 4

Reviewer 2

Reviewer 2 comment	Response location
General assessment	Combined Comments 3 and 16
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Comment 3	Combined Comment 9
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Comment 28	Combined Comment 8

Closing Statement

The revisions and explicit limitations provide a clearer account of the manuscript's evidentiary basis and practical interpretation.